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SIMATS ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

OBILI BHANU PRAKASH REDDY
192372075RA

QUESTIONS

- Q1 Library Book Management System 10 marks
- Q2 Student Grade Analyzer 1 mark
- Q3 Navigation History (Web Browser) 1 mark
- Q4 Employee Payroll System 1 mark

4/4 answered

Q1 Library Book Management System

10 marks

Problem Statement:

You are developing a Library Management System where books are stored in a HashSet to ensure no duplicate books (based on ISBN). The system needs to:

1. Add books with attributes: ISBN, title, author, and publication year
2. Remove a book by its ISBN
3. Display all books sorted by publication year (newest first)
4. Find all books by a specific author

Requirements:

- Use HashSet<Book> for storing books
- Implement Iterator to traverse and remove books
- Create a custom Comparator for sorting
- Display books using iterator

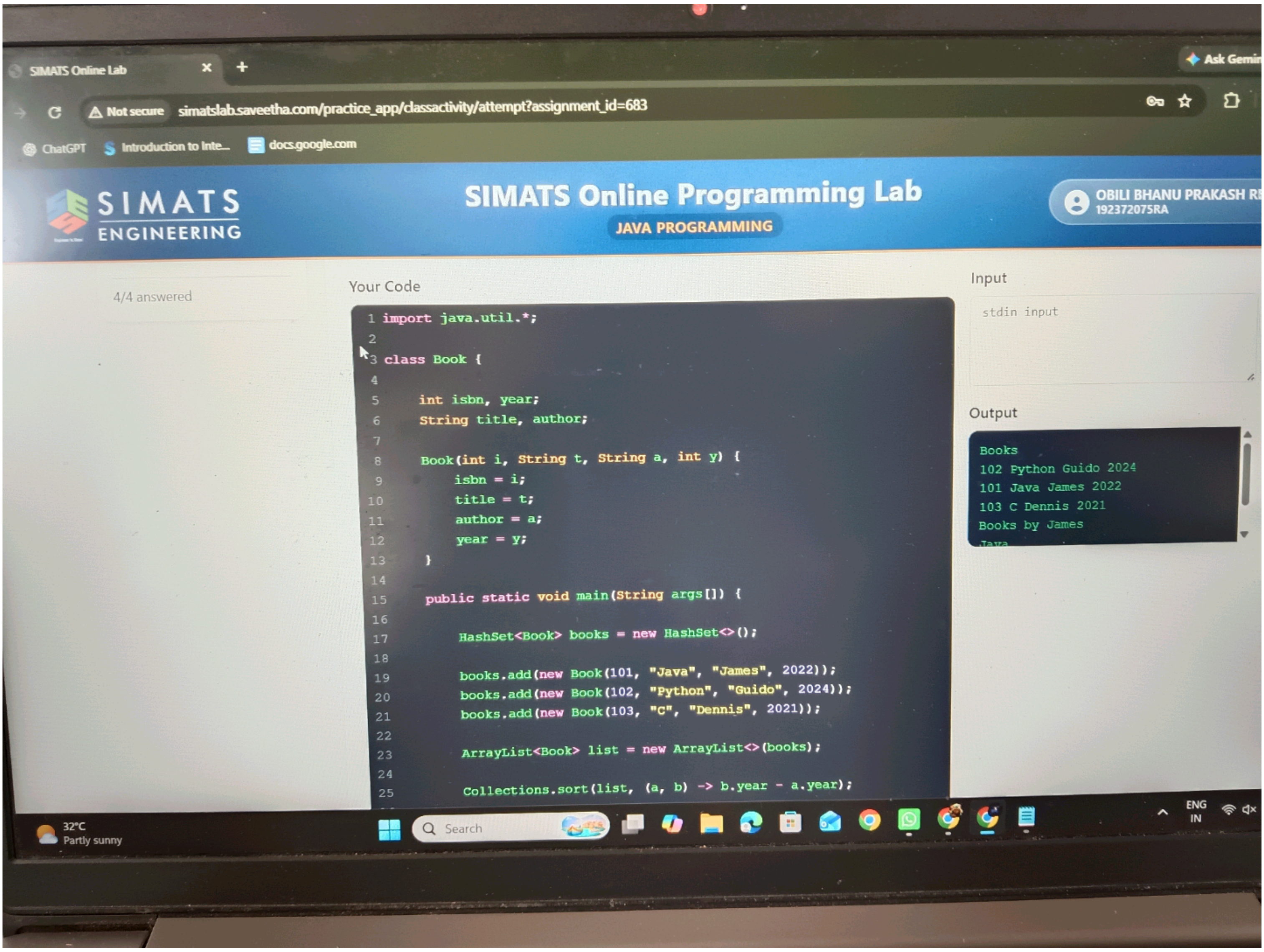
Your Code

```
import java.util.*;

class Book {

    int isbn, year;
    String title, author;

```

Input

stdin input

Output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4/4 answered
Your Code
import java.util.*;
class Book {
int isbn, year;
String title, author;
Book(int i, String t, String a, int y) {
isbn = i;
title = t;
author = a;
year = y;
}
public static void main(String args[]) {
HashSet<Book> books = new HashSet<>();
books.add(new Book(101, "Java", "James", 2022));
books.add(new Book(102, "Python", "Guido", 2024));
books.add(new Book(103, "C", "Dennis", 2021));
ArrayList<Book> list = new ArrayList<>(books);
Collections.sort(list, (a, b) -> b.year - a.year);
Input
stdin input
Output
Books
102 Python Guido 2024
101 Java James 2022
103 C Dennis 2021
Books by James
32°C
Partly sunny
Search
ENG
IN

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Collections (ALL)

Back

QUESTIONS

- Q1 Library Book Management System — 10 marks
- Q2 Student Grade Analyzer — 1 mark
- Q3 Navigation History (Web Browser) — 1 mark
- Q4 Employee Payroll System — 1 mark

4/4 answered

Q2 Student Grade Analyzer

1 mark

Problem Statement:

A school maintains student records with their grades in different subjects. You need to analyze and process this data using ArrayList and HashMap.

1. Store students in an ArrayList<Student> where Student has: ID, name, and a Map<String, Integer> (subject → grade)
2. Calculate the average grade for each student
3. Find the top 3 students with highest average grades
4. Remove students who have failed in more than 2 subjects (grade < 40)

Requirements:

- Use ArrayList<Student> for student list
- Use HashMap<String, Integer> for grades within each student
- Use Iterator to traverse and remove students
- Implement sorting using Collections.sort() with custom comparator

Your Code

```java
import java.util.*;

class Student {
```

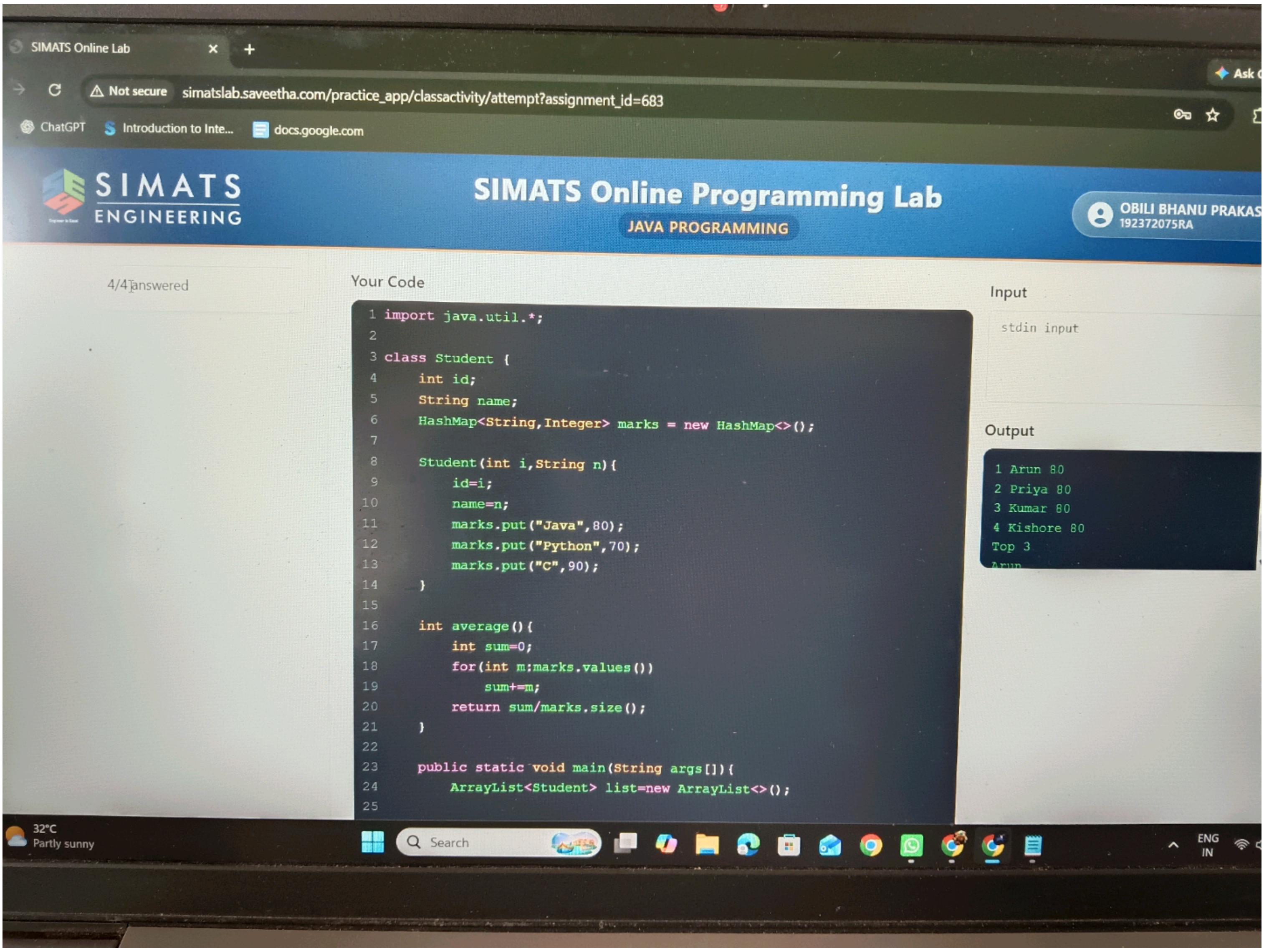
Input

stdin input

32°C
Partly sunny

Search

ENG
IN



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192372075RA

4/4 answered

Your Code

```
import java.util.*;

class Student {
    int id;
    String name;
    HashMap<String,Integer> marks = new HashMap<>();

    Student(int i,String n){
        id=i;
        name=n;
        marks.put("Java",80);
        marks.put("Python",70);
        marks.put("C",90);
    }

    int average(){
        int sum=0;
        for(int m:marks.values())
            sum+=m;
        return sum/marks.size();
    }

    public static void main(String args[]){
        ArrayList<Student> list=new ArrayList<>();

```

Input

stdin input

Output

```
1 Arun 80
2 Priya 80
3 Kumar 80
4 Kishore 80
Top 3
Arun
```

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Collections (ALL)

Back

QUESTIONS

Q1
Library Book Management System
10 marks

Q2
Student Grade Analyzer
1 mark

Q3
Navigation History (Web Browser)
1 mark

Q4
Employee Payroll System
1 mark

4/4 answered

Q3 Navigation History (Web Browser)

1 mark

Problem Statement:

You need to implement a web browser's navigation system where users can go back and forward through pages using a Stack.

1. Store visited URLs in a stack
2. Implement goBack() - moves one page back (but keeps forward history)
3. Implement goForward() - moves one page forward
4. Implement visitPage(String url) - adds a new page and clears forward history
5. Display current page and navigation history

Requirements:

- Use Stack<String> for back history
- Use Stack<String> for forward history
- Use Iterator to display all pages in both stacks without modifying them
- Handle edge cases (empty stack, end of navigation)

Your Code

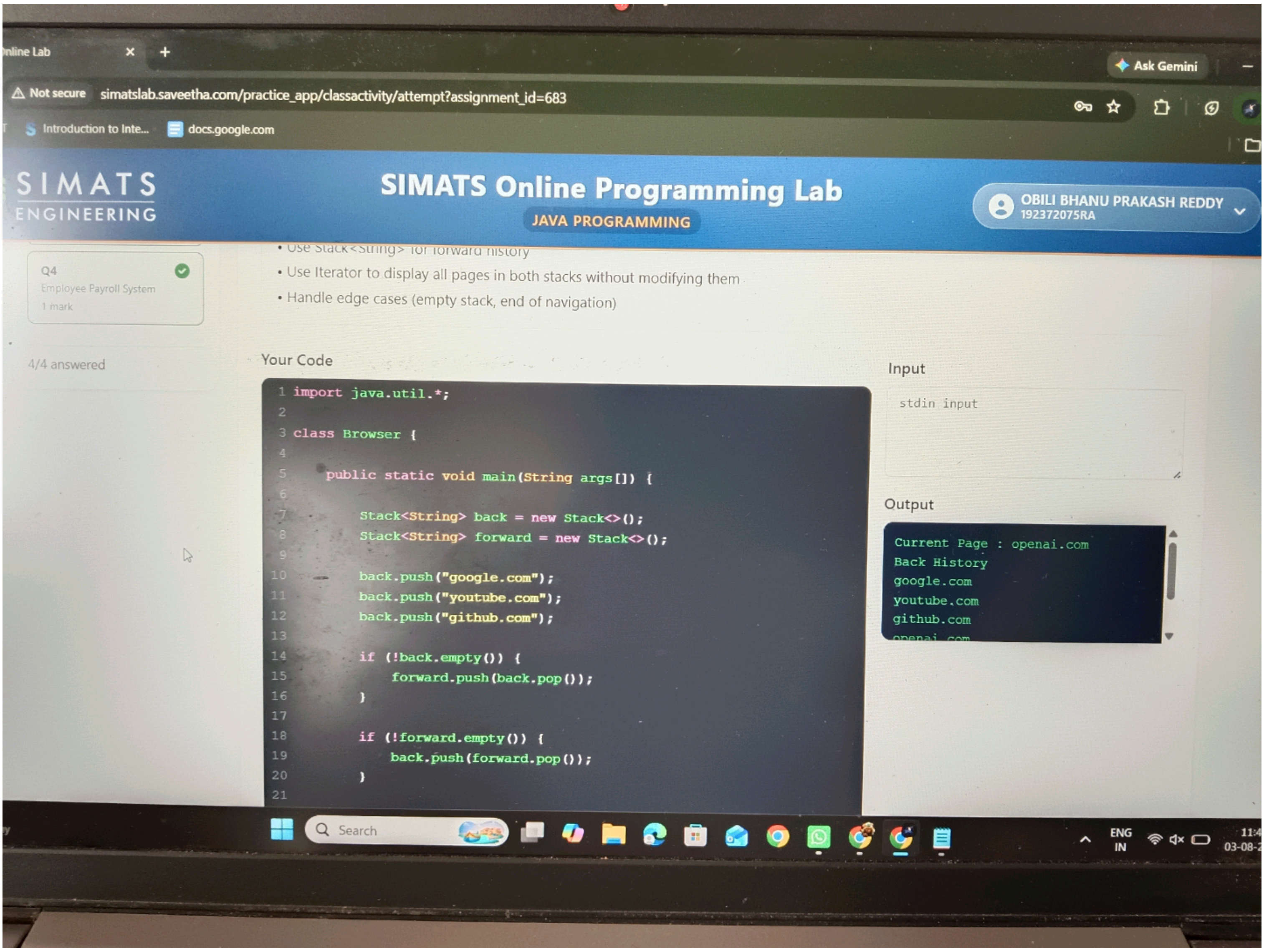
```
import java.util.*;

class Browser {
```

Input

stdin input

Search

ENG IN 11:45 03-08-20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Q4
Employee Payroll System
1 mark
4/4 answered
• Use Stack<String> for forward history
• Use Iterator to display all pages in both stacks without modifying them
• Handle edge cases (empty stack, end of navigation)
Your Code
import java.util.*;
class Browser {
public static void main(String args[]) {
Stack<String> back = new Stack<>();
Stack<String> forward = new Stack<>();
back.push("google.com");
back.push("youtube.com");
back.push("github.com");
if (!back.empty()) {
forward.push(back.pop());
}
if (!forward.empty()) {
back.push(forward.pop());
}
Input
stdin input
Output
Current Page : openai.com
Back History
google.com
youtube.com
github.com
Search
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Collections (ALL)

Back

QUESTIONS

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Library Book Management System
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Q2
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1 mark

Q3
Navigation History (Web Browser)
1 mark

Q4
Employee Payroll System
1 mark

4/4 answered

Q4 Employee Payroll System

1 mark

Problem Statement:

A company maintains employee records with their department and salary. You need to create a system that:

1. Store employees in a HashMap<String, List<Employee>> (Department → List of Employees)
2. Each Employee has: ID, name, department, salary
3. Calculate total salary expense per department
4. Find the highest-paid employee in each department
5. Remove employees with salary less than a threshold across all departments
6. Move an employee from one department to another

Requirements:

- Use HashMap<String, List<Employee>> for departmental storage
- Use Iterator to traverse both the map entries and the lists
- Use Iterator for safe removal while iterating
- Create methods to display department-wise statistics

Your Code

```
import java.util.*;
```

Input

stdin input

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```java
import java.util.*;

class Employee {
    int id, salary;
    String name, dept;

    Employee(int i, String n, String d, int s) {
        id = i;
        name = n;
        dept = d;
        salary = s;
    }

    public static void main(String args[]) {

        HashMap<String, List<Employee>> map = new HashMap<>();

        List<Employee> it = new ArrayList<>();
        it.add(new Employee(1, "Arun", "IT", 50000));
        it.add(new Employee(2, "Priya", "IT", 60000));

        List<Employee> hr = new ArrayList<>();
        hr.add(new Employee(3, "Kumar", "HR", 40000));
        hr.add(new Employee(4, "Anu", "HR", 55000));

        map.put("IT", it);
        map.put("HR", hr);
```

stdin input

Output

```
Department : HR
Total Salary : 95000
Highest Paid : Anu
Department : IT
Total Salary : 110000
Highest Paid : Priya
```